Program Name: A program to print 2D array in Java

Code:

```
public class printing2DMatrix {
    public static void main(String[] args){
        int[][] matrix = {{1,2,3},{4,5,6},{7,8,9}};
        for(int i=0; i< matrix.length; i++){
            for(int j=0; j<matrix[i].length; j++){
                System.out.print(matrix[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\
1 2 3
4 5 6
7 8 9

Process finished with exit code 0
```

Program Name: A program to transpose a matrix and print it in Java

Code:

```
public class printing2DTranspose {
    public static void printMatrix(int[][] matrix) {
        for (int[] row : matrix) {
            for (int element : row) {
                System.out.print(element + " ");
            }
            System.out.println();
        }
    }
    public static void main(String[] args) {
        int[][] matrix = {{1, 2, 3},{4, 5, 6},{7, 8, 9}};
        System.out.println("Original Matrix:");
        printMatrix(matrix);

        int rows = matrix.length, cols = matrix[0].length;
        int[][] transposed = new int[cols][rows];
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                transposed[j][i] = matrix[i][j];
            }
        }
        System.out.println("\nTransposed Matrix:");
        printMatrix(transposed);
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\
Original Matrix:
1 2 3
4 5 6
7 8 9

Transposed Matrix:
1 4 7
2 5 8
3 6 9

Process finished with exit code 0
```

Program Name: A program to print the Sum of Diagonal elements in Java

Code:

```
public class matrixDiagonalSum {
    public static void DiagonalSum(int[][] matrix){
    int pSum = 0, sSum = 0;
    int n = matrix.length;
    for(int i=0; i<n; i++){
        for(int j=0; j<n; j++){
            if(i==j) // Principle Diagonal
                pSum += matrix[i][j];
            if(i+j == n-1) // Secondary Diagonal
                sSum += matrix[i][j];
        }
    }
    System.out.println("Sum of Principle Diagonal " + pSum);
    System.out.println("Sum of Secondary Diagonal " + sSum);
    }
    public static void main(String[] args){
        int[][] matrix = {{2,4,5},{14,15,16},{33,56,78}};
        DiagonalSum(matrix);
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\
Sum of Principle Diagonal 95
Sum of Secondary Diagonal 53

Process finished with exit code 0
```

Program Name: A program to print the Triangular Sum of a matrix in Java

Code:

```java
public class matrixTriangularSum {
    public static void triangularSum(int[][] matrix){
        int upperSum = 0, lowerSum = 0;
        int n = matrix.length;
        for(int i=0; i<n; i++){
            for(int j=0; j<n; j++){
                if(i <= j) // Upper Triangular Sum
                    upperSum += matrix[i][j];
                if(j <= i) // Lower Triangular Sum
                    lowerSum += matrix[i][j];
            }
        }
        System.out.println("Sum of Upper Triangular " + upperSum);
        System.out.println("Sum of Lower Triangular " + lowerSum);
    }
    public static void main(String[] args){
        int[][] matrix = {{1,2,3},{4,5,6},{7,8,9}};
        triangularSum(matrix);
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\
Sum of Upper Triangular 26
Sum of Lower Triangular 34

Process finished with exit code 0
```